i When to Use

Use multiple regression when you have one quantitative criterion variable and two or more predictor variables (quantitative or qualitative). This allows you to evaluate the combined contribution of multiple predictors and identify which variables significantly contribute to explaining variation in the criterion.

H **Research Hypothesis:** Critics' ratings of Superman films are predicted by the actors' ages at the time of filming (Clark Kent and Lois Lane), the year the film was released, and the type of film (animated vs. live-action).

H **Null Hypothesis:** The combination of actor ages, year, and film type does not significantly predict critics' ratings.

Step 1: Examine Correlations Between Criterion and Predictors

Before fitting the regression model, examine the correlations between the criterion variable and each predictor. This helps you understand which variables are related to the outcome and can reveal potential suppressor effects (discussed later).

Correlations

		rt_critics_score	year	clark_age	lois_age
rt_critics_score	Pearson Correlation	1.000	−0.262	0.128	−0.109
	Sig. (2-tailed)		.530	.762	.798
	N	8	8	8	8
year	Pearson Correlation	−0.262	1.000	−0.234	0.498
	Sig. (2-tailed)	.530		.578	.209
	N	8	8	8	8
clark_age	Pearson Correlation	0.128	−0.234	1.000	0.449
	Sig. (2-tailed)	.762	.578		.265
	N	8	8	8	8
lois_age	Pearson Correlation	−0.109	0.498	0.449	1.000
	Sig. (2-tailed)	.798	.209	.265	
	N	8	8	8	8

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Notice the correlation pattern:

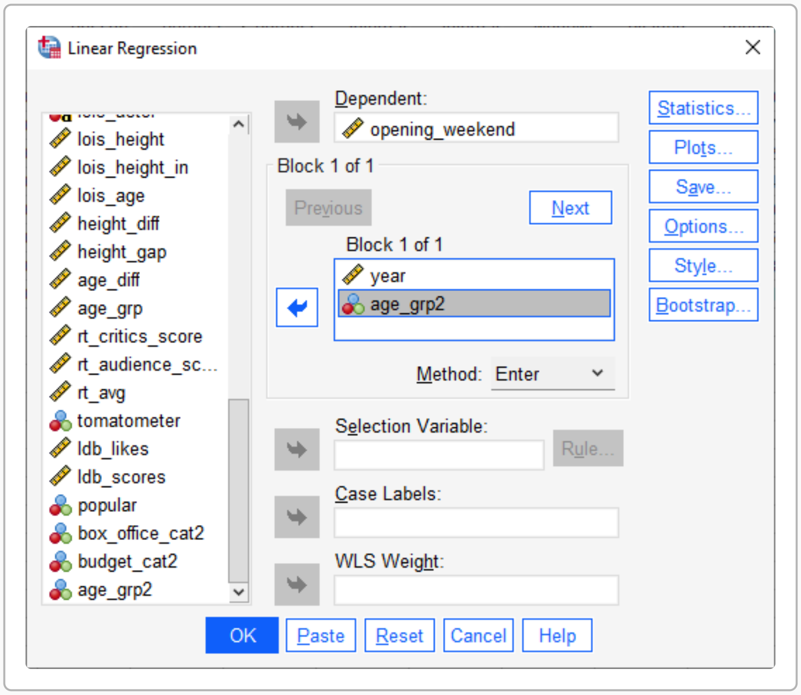
- **Year** and critics' score: $r = -0.262$
- **Clark Age** and critics' score: $r = 0.128$
- **Lois Age** and critics' score: $r = -0.109$

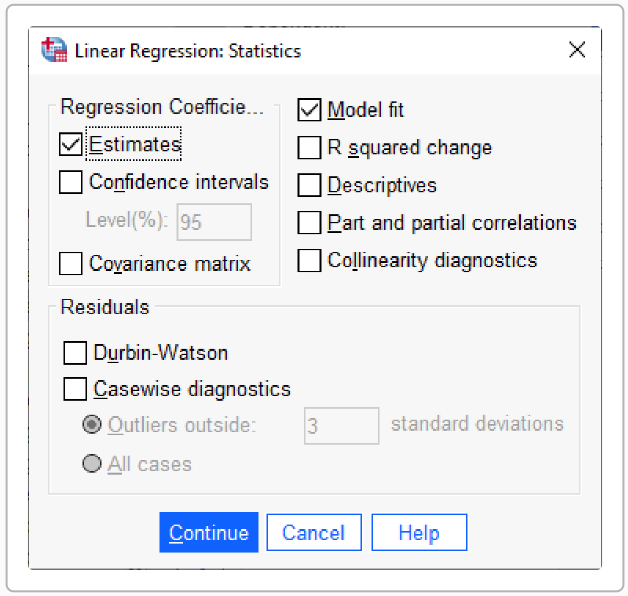
These bivariate correlations show which predictors are individually related to the criterion. In the multiple regression, some of these relationships may be modified by the presence of other predictors (indicating suppressor effects).

Step 2: Run the Multiple Regression

Analyze → Regression → Linear

- Move **rt_critics_score** to “Dependent”
- Move **year, clark_age, lois_age, type2** to “Independent(s)”
- Click “Statistics” — check:
  - Model fit (R^2, Adjusted R^2, F)
  - Estimates (b, Beta, t, Sig)
  - Confidence intervals for b
- Click “OK” to run the analysis





The order of entry for predictors does not matter in standard (simultaneous) regression. All predictors are evaluated in the context of all other predictors in the model.

</> SPSS Syntax

```
REGRESSION
  /MISSING=LISTWISE                              ①
  /STATISTICS=OUTS TOLS CONFINT                  ②
  /CRITERIA=PIN(.05) POUT(.10)                   ③
  /NOORIGIN
  /DEPENDENTrt_critics_score                     ④
  /METHOD=ENTER year clark_age lois_age type2.   ⑤
```

① Use listwise deletion for missing values
② Get outliers, tolerances, and confidence intervals
③ Set p-values for entry and removal
④ Specify the criterion variable
⑤ Enter all predictors simultaneously

SPSS Output

Model Summary

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	0.533	0.285	−0.669	13.602

ANOVA

Model	Sum of Squares	df	Mean Square	F	Sig.
Regression	220.830	4	55.208	0.298	.863
Residual	555.045	3	185.015		
Total	775.875	7			

Coefficients

	Unstandardized Coefficients		Standardized Coefficients		
Model	B	Std. Error	Beta	t	Sig.
(Constant)	310.162	586.415		0.529	.634
year	−0.128	0.293	−0.308	−0.438	.691
clark_age	0.469	1.663	0.193	0.282	.796
lois_age	−0.056	1.228	−0.035	−0.046	.967
type2	9.793	10.431	0.481	0.939	.417

The $\mathbf{R^2}$ value tells you the proportion of variance in the criterion that is explained by the model. Here, $R^2 = 0.285$, meaning the model explains approximately 28.5% of the variation in critics' scores.

The **ANOVA F-test** evaluates whether the overall model significantly predicts the criterion: - $F(4, 3) = 0.30$, $p = .863$

This indicates that the combination of predictors **does not significantly** predict critics' ratings.

Interpreting R^2

R^2 represents the **strength of the relationship** between the predictors and criterion:

- An R^2 of .25 suggests a **moderate** relationship
- An R^2 of .09 suggests a **weak** relationship
- An R^2 of .01 suggests a **very weak** relationship

The Adjusted R^2 penalizes for additional predictors, making it useful for comparing models with different numbers of predictors.

Individual Predictor Significance

The **Coefficients** table shows whether each individual predictor contributes significantly to the model. The t-test for each predictor tests:

H_0: This predictor does not significantly contribute to explaining the criterion ($b = 0$)

Unstandardized Coefficients (b)

- **year**: $b = -0.128$, $t(3) = -0.44$, $p = .691$
- **clark_age**: $b = 0.469$, $t(3) = 0.28$, $p = .796$
- **lois_age**: $b = -0.056$, $t(3) = -0.05$, $p = .967$
- **type2**: $b = 9.793$, $t(3) = 0.94$, $p = .417$

The **asterisk (*)** or "Sig." column indicates statistical significance: - $p < .05$: significant predictor - $p \geq .05$: not a significant predictor

A non-significant predictor is often removed to create a more parsimonious model.

Interpretation of Coefficients

Understanding b (Unstandardized) Weights

The **unstandardized coefficient (b)** represents the change in the criterion for a one-unit increase in that predictor, **holding all other predictors constant**.

For example, if b for year = −0.1283:

- For each additional year of film release (holding actor ages and type constant), the predicted critics' score **changes by −0.1283 points**

Suppressor Variables

Sometimes a predictor's **b weight has the opposite sign from its correlation with the criterion**. This occurs when the predictor explains unique variance while controlling for other predictors.

- The **sign** of b indicates direction: positive (increases) or negative (decreases)
- The **magnitude** of b indicates how much the criterion changes

This is not "suppression" in the causal sense — it's simply the predictor's contribution after accounting for multicollinearity.

Understanding Standardized (Beta) Weights

The **standardized coefficient (Beta)** converts b to a common scale (standard deviation units). This allows direct comparison of which predictors have the strongest unique contribution to the model.

For example, if Beta for year = 0.35 and Beta for clark_age = 0.18: - Year has a stronger unique effect than Clark's age - The relative magnitude of Beta values shows comparative importance

Step 3: Applying the Regression Model

Once you have a significant model with significant predictors, you can use it to compute **predicted criterion scores** for new cases (or existing cases, to examine residuals).

Using Transform → Compute to Calculate Predicted Values

In SPSS, you would create a new variable using the unstandardized coefficients:

</> **SPSS Syntax**

```
COMPUTE predicted_score =
  intercept_value
  + (b_year * year)
  + (b_clark * clark_age)
  + (b_lois * lois_age)
  + (b_type * type2).
EXECUTE.
```

1. Replace intercept_value, b_year, etc. with actual values from your output
2. List each predictor multiplied by its unstandardized coefficient
3. Add all terms together to get the predicted score
4. EXECUTE to apply the formula to all cases

For our example, using the coefficients:

```
COMPUTE predicted_score =
  310.16
  + (-0.1283 * year)
  + (0.4695 * clark_age)
  + (-0.0559 * lois_age)
  + (9.7935 * type2).
EXECUTE.
```

The predicted scores are the model's best estimates of what the criterion value should be for each case, based on their predictor values. Differences between predicted and actual values (residuals) indicate how well the model fits each case.

Considerations When Using Predictions

When using regression predictions, keep these important limitations in mind:

1. **Model Fit**: Predictions are only as good as the model. An R^2 of .40 means 60% of variance is unexplained. Use the **Standard Error of the Estimate (SEE)** to construct confidence intervals around predictions.

2. **Extrapolation**: Do not use the regression model to make predictions outside the range of your original data. The relationship may not hold beyond observed values.

3. **Sample Size**: Predictions are less stable with small samples. Larger, more representative samples produce more generalizable predictions.

4. **Assumption Violations**: If regression assumptions (linearity, homoscedasticity, normality of residuals) are violated, your predictions may be biased.

Example Write-Up

¶ Sample APA Write-Up

A multiple regression model was fit to examine whether critics' ratings of Superman portrayals (*M* = 79.38, *SD* = 10.53) were predicted by release year (*M* = 1998.12, *SD* = 25.30), Clark Kent actor age (*M* = 29.67, *SD* = 4.33), Lois Lane actress age (*M* = 30.84, *SD* = 6.50), film type (*M* = 1.38, *SD* = 0.52). The overall model was not significant, $F(4,3) = 0.30$, $p = .863$, explaining 28.5% of the variance in critics' ratings ($R^2 = 0.285$). The following predictors were not significant: release year (*b* = −0.1283, *Beta* = −0.308, *t* = −0.44, *p* = .691); Clark Kent actor age (*b* = 0.4695, *Beta* = 0.193, *t* = 0.28, *p* = .796); Lois Lane actress age (*b* = −0.0559, *Beta* = −0.035, *t* = -0.05, *p* = .967); film type (*b* = 9.7935, *Beta* = 0.481, *t* = 0.94, *p* = .417).

Summary Table

Typically, results are presented in a summary table showing descriptive statistics, intercorrelations, and regression coefficients:

Predictor	*B*	*SE*	*β*	*t*	*p*
Year	-0.13	0.29	-.31	-0.44	.691
Clark Age	0.47	1.66	.19	0.28	.796
Lois Age	-0.06	1.23	-.03	-0.05	.967
Type	9.79	10.43	.48	0.94	.417